B NANDIVARDHAN REDDY

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Career Objective

Embedded Systems and IoT enthusiast with hands-on experience in STM32, ESP32, and hardware interfacing. Passionate about developing intelligent embedded solutions involving firmware development, sensor integration, and real-time monitoring systems. Seeking an internship opportunity to contribute to innovative embedded and IoT engineering solutions.

Education

B.Tech in Electronics and Communication Engineering 2024 – 2027
KL Deemed to be University, Guntur
CGPA: 8.45

Diploma in Electronics and Communication Engineering 2021 – 2024
Kuppam Engineering College, Chittoor
Percentage: 73%

Technical Skills

Programming:	C, Embedded C, Python
Microcontrollers:	STM32, ESP32, Raspberry Pi, Arduino
Protocols:	UART, I2C, SPI, CAN
Tools:	STM32CubeIDE, Keil, Arduino IDE, EasyEDA
Embedded Concepts:	Sensor Interfacing, PCB Basics, Firmware Development, Hardware Debugging

Internship Experience

Junior Quality Inspector — SMILE Electronics Limited, Bengaluru Jan 2024 – Jun 2024

- Inspected PCB assemblies and verified soldering quality and component placement.
- Performed visual inspections and functional testing to ensure product reliability and compliance with manufacturing standards.
- Assisted in defect analysis, rework coordination, and maintaining quality inspection records.

Projects

Vehicle-to-Vehicle (V2V) Communication System using ESP32

- Developed an ESP32-based V2V communication system for real-time exchange of vehicle safety information.
- Implemented wireless communication for collision warning and emergency alert transmission between vehicles.
- Designed system architecture for intelligent transportation and road safety applications.

AI-Enabled Smart Battery Swapping Hub for Electric Vehicles

- Developed an ESP32-based EV battery swapping system with NFC authentication and Firebase cloud integration.
- Implemented real-time monitoring, user logging, and machine learning-based battery health prediction.
- Designed automated locking mechanism for secure battery access control.

Earthquake Monitoring System using ESP32

- Engineered earthquake monitoring system using ESP32 and MPU6050 accelerometer for vibration detection.
- Implemented multi-level alert system using OLED display, buzzer, and SMS notifications via Twilio API.

Certifications

- Altium 365 Design Management
- Embedded System Application and IoT Programming – L1
- IIoT Programming & Automation – L2

Technical Events & Participation

SDG Innovation Challenge — Techno Summit 2025, Sathyabama Institute of Science and Technology (Project Demonstration Participant)